

ECON 282E — Homework 1, Part A

Dynamic Programming on the Computer

Foundations of Macroeconomics
University of California, Riverside

Out: Thursday, October 8, 2026 • Due: Thursday, October 22, 2026

What to hand in. One PDF with your answers and figures, the code that produced them, and an `AI_LOG.md` recording how you used any AI assistance. Part B is released on October 15 and is due the same day as this part.

On the computational questions. Every number you report must come from code you ran. State the grid, the tolerance and the dtype alongside each result — the reporting standard from Session 3. Question 2(c) uses techniques taught on **October 15**; you can do 2(a) and 2(b) as soon as this sheet is out.

AI policy. Assistance is allowed and expected. You remain responsible for every number. The validation is the part you cannot delegate.

Question 1. The model on paper (30 points)

A representative household solves

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \ln c_t \quad \text{s.t.} \quad c_t + k_{t+1} = Zk_t^\alpha, \quad c_t \geq 0, \quad k_{t+1} \geq 0, \quad k_0 > 0,$$

with $\alpha \in (0, 1)$, $\beta \in (0, 1)$, $Z > 0$, and **full depreciation**, $\delta = 1$. Throughout Part A use $\alpha = 0.36$, $\beta = 0.95$ and $Z = 1$.

- (a) **(5 pts)** Write the sequence problem and the corresponding Bellman equation. State the state variable, the control, and the feasible correspondence $\Gamma(k)$.
- (b) **(8 pts)** Verify Blackwell's sufficient conditions — monotonicity and discounting — for the operator T . What do they, together with the contraction mapping theorem, allow you to conclude about existence, uniqueness and convergence?
- (c) **(10 pts)** Guess $v(k) = A + B \ln k$ and verify it. Solve for A and B , and show that the optimal policy is $k' = g(k) = \alpha\beta Zk^\alpha$ and consumption is $c = (1 - \alpha\beta)Zk^\alpha$. Report A and B numerically at the calibration above.
- (d) **(7 pts)** The contraction theorem gives two different bounds. Using $\|v_n - v^*\| \leq \frac{\beta}{1-\beta} \|v_n - v_{n-1}\|$, answer:
 - (i) If you stop when successive iterates differ by $\varepsilon = 10^{-6}$, how far from v^* might you still be, at $\beta = 0.95$? At $\beta = 0.99$?
 - (ii) Using $n \gtrsim \ln \varepsilon / \ln \beta$, how many sweeps does $\varepsilon = 10^{-6}$ require at each of those two discount factors?

Comment on what this implies for reporting a “converged” solution.

Question 2. The model on the computer, parts (a)–(c) (70 points)

Solve the same model numerically. Use a capital grid on $[0.5k^*, 1.5k^*]$ where $k^* = (\alpha\beta Z)^{1/(1-\alpha)}$ is the deterministic steady state, and `float64` throughout.

- (a) **(25 pts) Grid value function iteration.** Implement VFI with the choice k' restricted to the grid (no interpolation, no continuous maximization). With $N = 200$ and tolerance 10^{-6} , report:
- (i) the number of sweeps, and how it compares with your answer to 1(d)(ii);
 - (ii) $\max_k |v(k) - v^*(k)|$ and the maximum *relative* policy error $\max_k |g(k)/g^*(k) - 1|$, against the closed form of 1(c);
 - (iii) a plot of the computed policy against $\alpha\beta Z k^\alpha$, zoomed into a window of about 30 consecutive grid points.

What does the zoomed plot show, and why does it not improve with more sweeps?

- (b) **(25 pts) Refinement, and the error budget.** Solve at $N \in \{50, 100, 200, 400, 800\}$ and tabulate the value and policy errors against h . Fit the slopes of $\log(\text{error})$ against $\log h$.
- (i) What slopes do you get, and what do Santos & Vigo-Aguiar (1998) predict?
 - (ii) Repeat with tolerance 10^{-11} . One of the two slopes changes materially. Which, and why? Relate your answer to 1(d)(i).
- (c) **(20 pts) Acceleration.** (*Techniques from Session 4, October 15.*) Starting from your (a) solver, add each of the following and report a table of policy updates, wall time and the maximum relative policy error:
- (i) Howard's policy improvement with $n_H \in \{10, 20, 50\}$;
 - (ii) exploiting monotonicity of g to restrict the search;
 - (iii) both together.

Two things to comment on: which of these changes the *answer* and which only changes the route to it; and whether your wall-clock times move in the same direction as your operation counts. Explain any disagreement.

Reminder. Report the grid, the tolerance, the dtype and the hardware with every timing. An accuracy claim without a benchmark, and a speed claim without a timing, both count as unsupported.